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ECEN 3730 – PCB Lab

Lab 05 Report

09/14/25

PCB Lab 05 Report

Objective:

This lab focuses on measuring the outputs created a MOSFET circuit to understand switching noise. The goal is to understand why a decoupling capacitor works and how it changes the measured wave depending on its capacitance and its location within the circuit.

Equivalent Circuit:

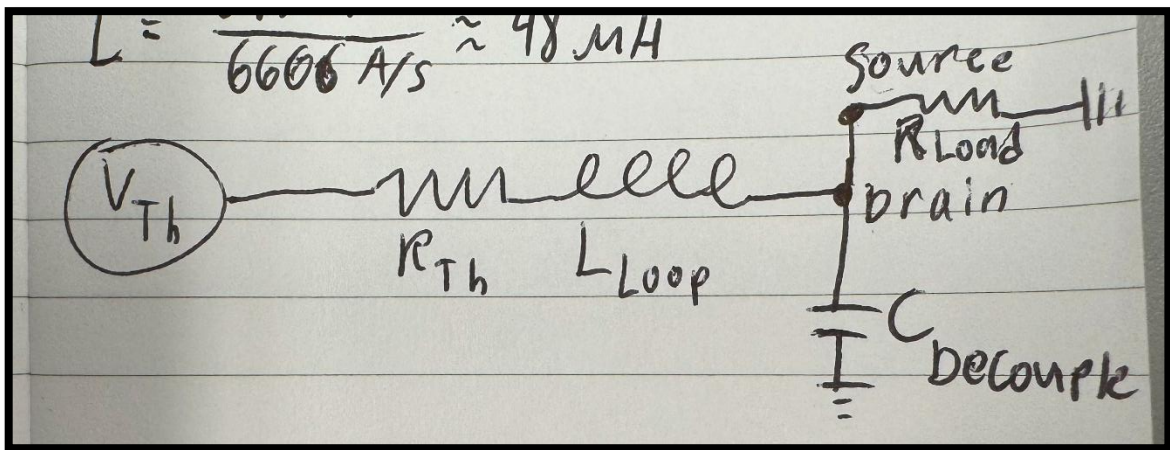


Figure 1: Equivalent Circuit

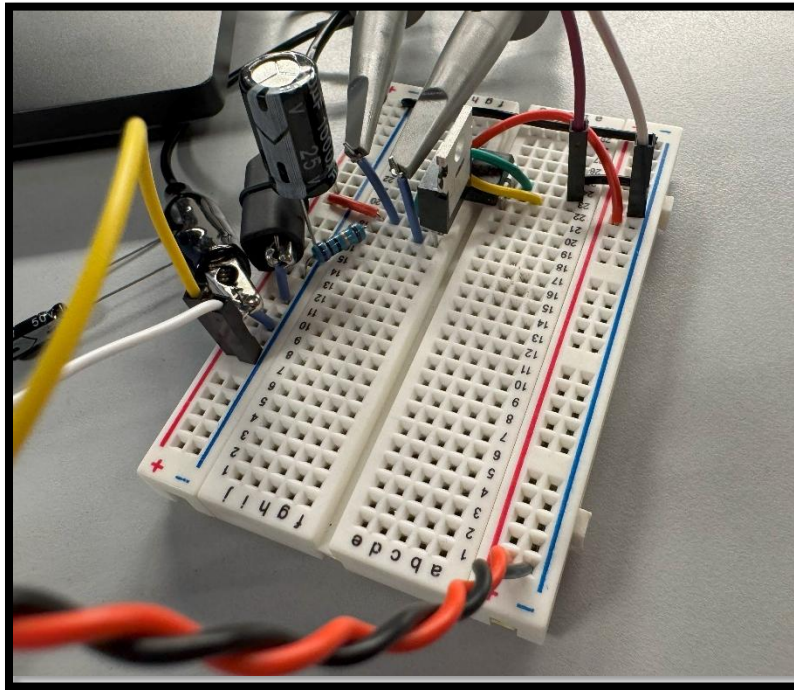
Circuit Tested:

Figure 2: Final Measured Circuit with 1000 μ F Decoupling Capacitor on High Switch

Oscilloscope Measurements:

The following oscilloscope captures are measurements of the above circuit set to the low switch position. Three measurements were recorded. Without a decoupling capacitor, with a 1 μ F decoupling capacitor, and with 1000 μ F decoupling capacitor. This allows for an easier understanding of the effects of a decoupling capacitor by viewing the oscilloscope capture. The first figure shows one cycle of the output wave of the source and drain without a decoupling capacitor. The source is the yellow wave and the drain is the purple wave.

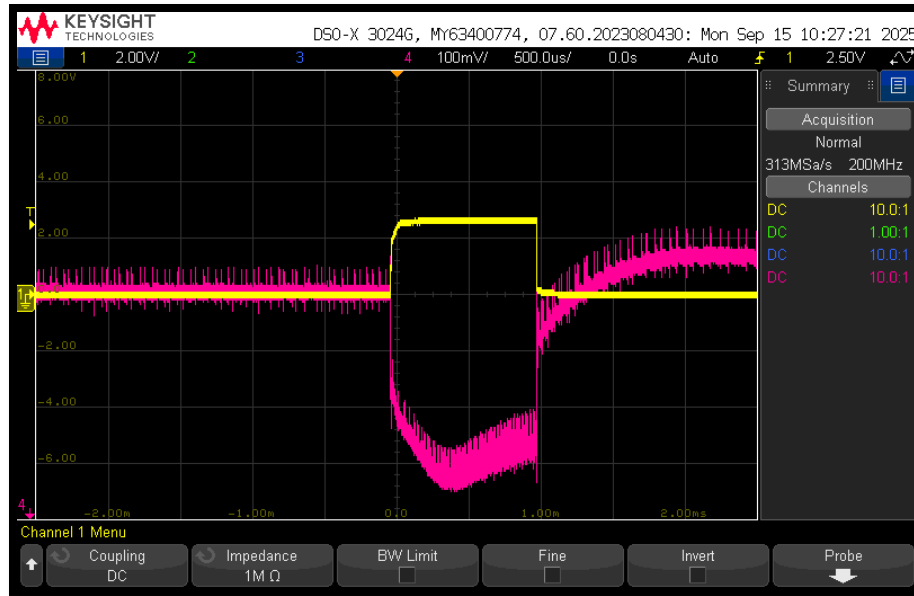


Figure 3: Full View of the Source and Drain Wave. No Decoupling Capacitor

This view provides an idea of what one cycle of the wave without a decoupling capacitor looks like. The noise on the purple drain wave is apparent. To make this clearer, however, the following measurements observe the rising edge of the waves. The yellow depicts the source, the purple depicts the newly measured wave, and the orange depicts the previously measured noise.

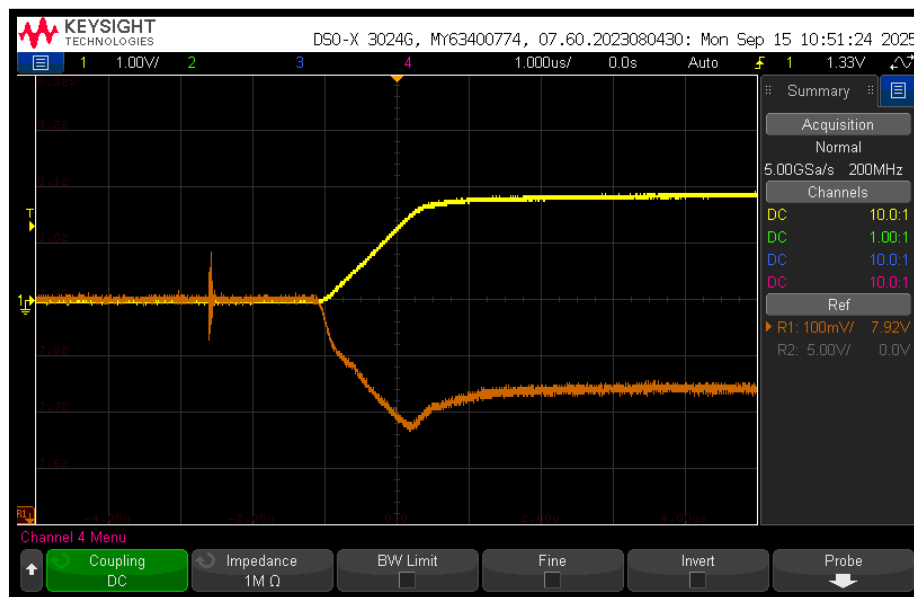


Figure 4: Rising Edge of the Source and Drain Wave. No Decoupling Capacitor

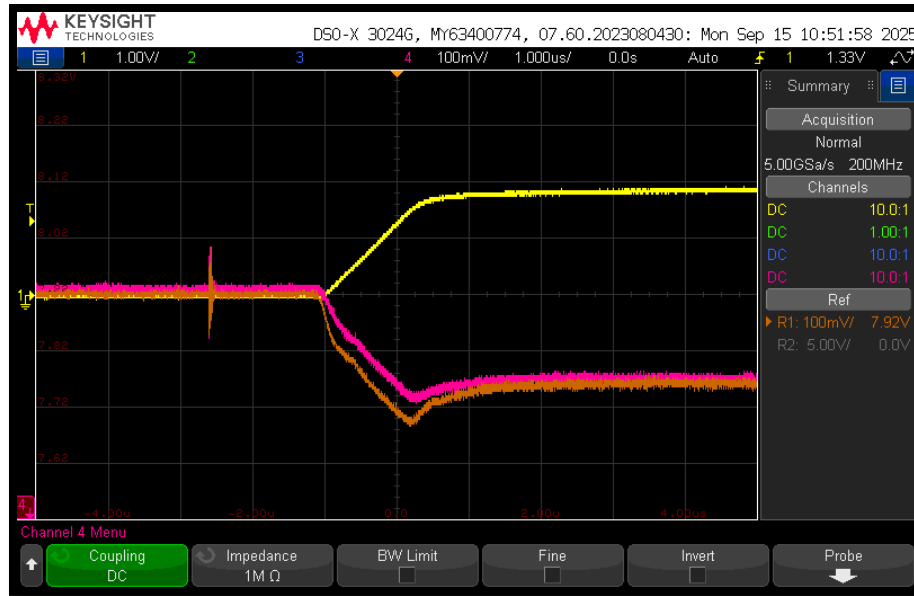


Figure 5: Rising Edge of the Source and Drain Wave. $1\mu\text{F}$ Decoupling Capacitor

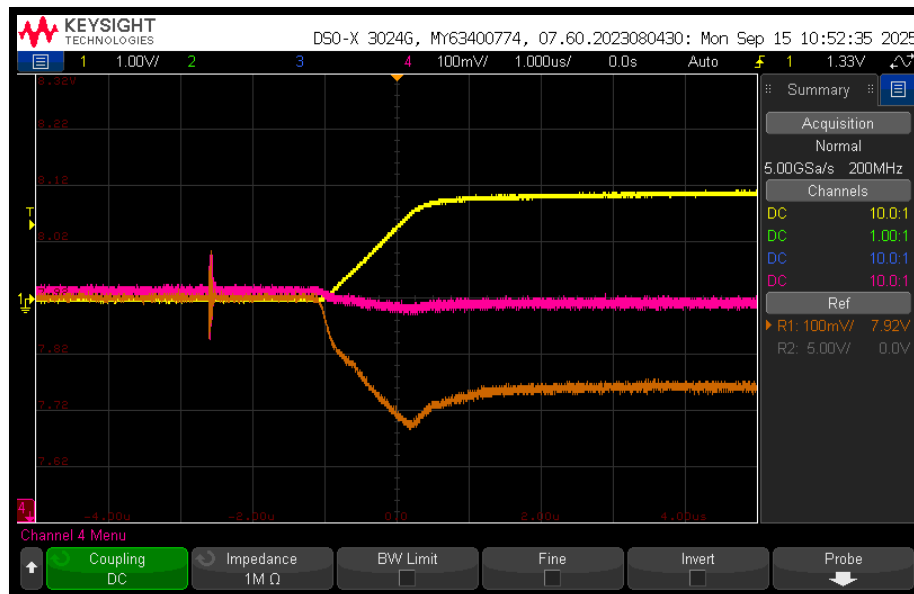


Figure 6: Rising Edge of the Source and Drain Wave. $1000\mu\text{F}$ Decoupling Capacitor

It is easy to observe the effect a decoupling capacitor has when able to compare the new results to the initially measured noise. *Figure 3* shows the measured noise on the rising edge of the slow switch with no decoupling capacitor. It deviates significantly from the source wave. *Figure 4* and *Figure 5* illustrate how a decoupling capacitor reduces the noise and brings the

drain wave closer to the source. It is apparent that the 1000 μ F decoupling capacitor has a far greater change on the output compared to the 1 μ F decoupling capacitor. The next measurement determines how the position of the capacitor affects the noise.

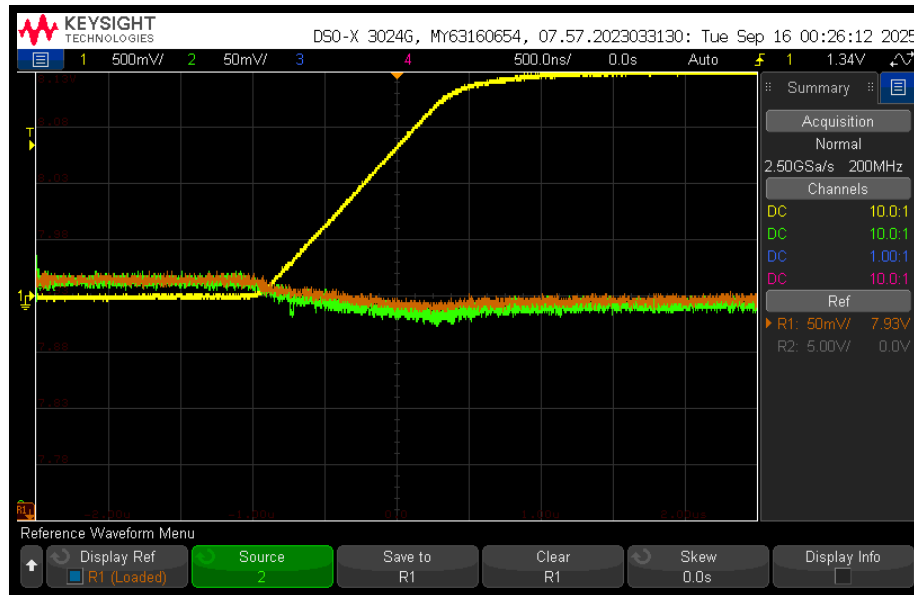


Figure 7: Rising Edge of the Source and Drain Wave. Different 1000 μ F Decoupling Capacitor Positions

The same rising edge is measured, but this time at 500 ns. The yellow line still depicts the source wave, the orange wave depicts the 1000 μ F decoupling capacitor closer to the circuit, and the green wave depicts the 1000 μ F capacitor closer 9V supply. Although slight, the capacitor is more effective the closer it is to the circuit than being near to the supply.

Calculations:

$$R_{Th} = R_{Load} \frac{(V_{Th} - V_{Load})}{V_{Load}}$$

$$R_{Th} = (10\Omega) \frac{(8V - 7.68V)}{7.68V} = 0.416\Omega$$

$$\Delta V = L \frac{dI}{dt}$$

$$L = (0.32V) \left(6666.67 \frac{A}{s} \right) = 48\mu H$$

Concluding Thoughts:

I have learned to power a circuit and utilize a Thevenin model with series resistance and loop inductance. I know how to look at a circuit with a different point of view and understand what will be affected when a component changes. By taking data between the VRM and through the MOSFET source and drain, I can calculate the behavior of the circuit depending on if it is positioned to fast or slow. Taking more precise measurements would have been beneficial which I now have learned for next time. Decoupling capacitors are effective at reducing the observed switching noise, but their capacitance and position in the circuit still contribute to this.